

Palak Shah

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SUMMARY

Software Engineer with extensive experience in machine learning, ranking algorithms, and Gen AI development within global-scale environments (Maps, Auto, Ads). Committed to high-quality engineering standards and architectural integrity. My work is defined by a relentless curiosity and a commitment to technical excellence; I believe in building technology that isn't just powerful, but purposefully designed to solve real-world complexities.

EXPERIENCE

Google, New York — Software Engineer

July 2024 - November 2025

AI Mode (Backend / Gen AI)

Google's AI-Mode serves as an intelligent conversational layer, transforming standard search into an intuitive, multi-modal query experience. For this project, I designed and deployed a hierarchical AI framework for local query resolution.

- Built a model to understand user intent and generate high-quality, complementary fan-out queries.
- Architected the interface between generative models and legacy Maps backend services to ensure low-latency response.
- Trained a summarization model to aggregate backend outputs.
- Used auto-raters to measure the quality in terms of helpfulness and correctness of the results.

Mar 2022 - June 2024

Maps Quality (Backend / Ranking)

Google Maps features a "Search Along Route" (and the more recent "Explore Along Your Route") functionality designed to help drivers find stops—such as gas stations, restaurants, or scenic viewpoints—without significant detours from their planned path.

- Engineered high-throughput data ingestion and transformation pipelines to power ranking models.
- Leveraged deep neural network architectures to refine Maps ranking algorithms, reducing latency for Google's Search Along Route feature.

Oct 2023 - Feb 2024

Maps Quality (Backend / Ranking)

Article 5(2) of the Digital Markets Act (DMA) is designed to stop large tech companies from creating an unfair data advantage by automatically

EDUCATION

MS in Electrical and Computer Engineering

University of Florida

August 2013 - May 2015

BTech in Information and Communication Technology

Dhirubhai Ambani Institute

August 2009 - May 2013

SKILLS

Languages : C++, Java, Python, SQL, GCL

Tools : VSCode, Kiro, Google Colabs, Jupyter, Google Web Toolkit (GWT), Git, Vim

Database Technologies :
Google's Bigtable, Spanner, Flume, MapReduce; Apache's Storm, Kafka;

CONNECT

<https://www.linkedin.com/in/palakshah2015/>

pooling your information across all their services. I was part of a rapid response team that made sure Maps ranking pipelines were DMA 5(2) compliant.

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January 2020 - March 2022

Google Assistant, Android Auto (Backend / On-device)

Android Automotive OS (AAOS) is an operating system built directly into the car. Unlike Android Auto, it does not require a smartphone to run.

- Updated on-device Natural Language Processing models to improve understanding of directional and local search queries.
- Optimized on-device server infrastructure to efficiently query locally stored maps, enabling the user to search while offline.

Google, California — Software Engineer

July 2016 - December 2019

Ads Privacy and Security (Backend)

Founding member of the Ads Integrity: Human Interactions team, focused on bridging the gap between platform security and advertiser productivity by assisting in the creation of policy compliant ads.

- Led the migration of large-scale advertising datasets to Bigtable, architecting distributed Flume/MapReduce pipelines to ensure data integrity.
- Designed and owned a high-throughput RPC-based rule engine serving as the core policy enforcement layer across Google Ads, AdWords Express, and the Ads API.
- Engineered real-time advertiser messaging and automated re-crawl/re-review pipelines. These systems streamlined creative compliance workflows, significantly reducing latency and operational overhead in labeling queues.
- Set up rigorous metrics collection and performance monitoring via alerting. Managed high-pressure on-call rotations to maintain strict SLOs.

July 2018 - December 2019

Ads Privacy and Security (Community Contributions)

Engineering Excellence is the intentional, disciplined practice of building systems that are not just functional, but also sustainable, scalable, and maintainable. It is measured through code quality, tech debt reduction and comprehensive testing.

- Headed a cross-functional initiative focused on Engineering Excellence, driving measurable improvements in code quality, technical debt reduction, and team velocity over multiple quarters.

Google, New York— *Engineering Resident*

July 2015 - July 2016

Google's Engineering Residency is an early-career program focused on practical, hands-on experience, and mentorship to transition college graduates into full-time software engineers.

Hiring Applications (Full Stack)

- Productionized Google's innovative hiring software using Google Web Toolkit (GWT), CSS and HTML.
- Delivered scalable front-end solutions that enhanced the efficiency of hiring pipelines while maintaining rigorous code quality and accessibility standards.

